1 Mask

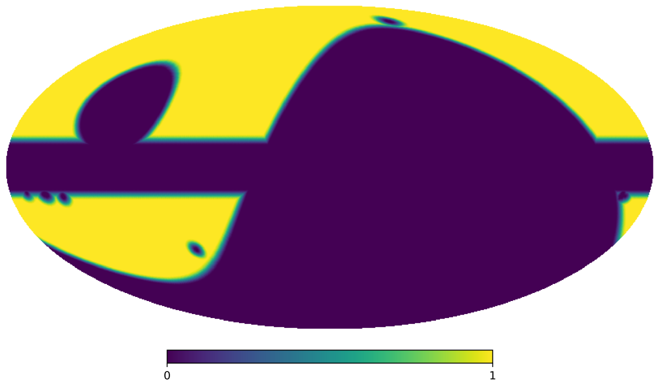

Figure 1: Analysis mask combining the QUIJOTE low-declination survey mask with a 10 mK intensity threshold applied to the QUIJOTE 11 GHz map. The mask excludes regions with high foreground contamination and point sources to ensure reliable polarimetric analysis of diffuse Galactic emission.

2 Data

Our analysis employs two complementary datasets: simulated sky maps generated using the `Python` Sky Model (`PySM`) library, which incorporates all major Galactic foreground components; and observational data from the QUIJOTE, WMAP, and Planck experiments. All simulated maps are convolved with the appropriate beam window functions for each respective experiment to ensure realistic instrumental effects.

Before fitting the observed power spectra, we subtract the CMB contribution using the best-fit theoretical ΛCDM spectrum from Planck 2018 (Planck Collaboration, 2020). This spectrum is obtained from the full Plik TTTEEE likelihood combined with low-multipole (lowE) and gravitational lensing constraints. For polarization modes, we subtract both C_ℓ^{EE} and C_ℓ^{BB}, where the latter includes only the contribution from B-modes generated by gravitational lensing, since primordial B-modes of inflationary origin have not yet been detected. This CMB subtraction is essential to isolate the Galactic foreground emission (synchrotron and thermal dust) that constitutes the target of this analysis.

3 Error estimation

Power spectrum uncertainties are estimated through Monte Carlo simulations for both datasets. We compute the standard deviation from 100 realizations each of sky+noise and pure noise maps. The noise characteristics vary by experiment: QUIJOTE and Planck employ the official noise simulations provided by their respective collaborations, incorporating both white noise and correlated $1/f$ noise components. For WMAP, we model the noise as purely white, following standard practice for this dataset.

4 MCMC fitting

The total angular power spectrum between frequencies ν_i and ν_j is modeled as

$$C_\ell(\nu_i,\nu_j) = C_\ell^{\rm sync}(\nu_i,\nu_j)+C_\ell^{\rm dust}(\nu_i,\nu_j)+C_\ell^{\rm sd}(\nu_i,\nu_j), \tag{1}$$

where the synchrotron, dust, and synchrotron-dust cross-correlation terms are

$$C_\ell^{\rm sync}(\nu_i,\nu_j) = A_s\left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_s}\left(\frac{\nu_i}{\nu_{\rm ref}}\right)^{\beta_s}\left(\frac{\nu_j}{\nu_{\rm ref}}\right)^{\beta_s}, \tag{2}$$

$$C_\ell^{\rm dust}(\nu_i,\nu_j) = A_d\left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_d} S_d(\nu_i)S_d(\nu_j), \tag{3}$$

$$\begin{aligned} C_\ell^{\rm sd}(\nu_i,\nu_j) = \rho\sqrt{A_sA_d}\Big[&S_s(\nu_i)S_d(\nu_j) \\ &+S_s(\nu_j)S_d(\nu_i)\Big]\left(\frac{\ell}{\ell_{\rm ref}}\right)^{(\alpha_s+\alpha_d)/2}, \end{aligned} \tag{4}$$

with $S_s(\nu) = (\nu/\nu_{\rm ref})^{\beta_s}$ and the modified blackbody scaling in Rayleigh-Jeans units

$$S_d(\nu) = \frac{B_\nu(\nu,T_d)}{B_\nu(\nu_0,T_d)}\left(\frac{\nu}{\nu_0}\right)^{\beta_d}, \tag{5}$$

where B_ν is the Planck function. Here, A_s and A_d are the amplitudes of the synchrotron and dust components at the reference multipole $\ell_{\rm ref} = 80$ and frequencies $\nu_{\rm ref} = 11.1$ GHz and $\nu_0 = 353.0$ GHz, respectively; α_s and α_d are the multipole spectral indices; β_s and β_d are the frequency spectral indices; T_d is the dust temperature (fixed at $T_d = 19.6$ K); and ρ is the synchrotron-dust correlation coefficient.

4.1 Data Selection and Frequency Coverage

We perform a Markov Chain Monte Carlo (MCMC) fitting procedure using data from QUIJOTE 11 GHz; WMAP 23, 33, 41, 61, and 94 GHz; and Planck 30, 44, 70, 100, 143, 217, and 353 GHz, covering the multipole range $30 < \ell < 200$. To avoid noise bias in the polarization power spectra, we exclusively use cross-spectra between different frequency bands ($i \neq j$), discarding all auto-spectra ($i = j$). This yields 78 independent cross-spectra from the 13 frequency bands, which improves the robustness of the fitted spectral indices while mitigating systematic effects from instrumental noise.

4.2 Bayesian Inference

We adopt a Bayesian framework to constrain the model parameters $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$. The posterior distribution is given by

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{6}$$

where $\mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior. The likelihood is constructed from the observed power spectra $\hat{C}_\ell(\nu_i, \nu_j)$ and their uncertainties:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i,j,\ell} \frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \tag{7}$$

For the power-law model fitting, we adopt uniform (flat) priors for all model parameters within physically motivated bounds:

$$A_s \in [0, \infty), \tag{8}$$
$$\alpha_s \in [-6, 0], \tag{9}$$
$$\beta_s \in [-6, 0], \tag{10}$$
$$A_d \in [0, \infty), \tag{11}$$
$$\alpha_d \in [-6, 0], \tag{12}$$
$$\beta_d \in [0, 6], \tag{13}$$
$$\rho \in [-1, 1], \tag{14}$$

For the bin-to-bin analysis, to regularize the fit and incorporate external information from the literature, we impose a Gaussian prior on the synchrotron frequency spectral index:

$$\beta_s \sim \mathcal{N}(-3.1, 0.30^2), \tag{15}$$

while maintaining uniform priors for the remaining parameters within the bounds specified above.

5 Results

5.1 Simulated data

We compute the angular power spectra from the PySM simulated maps with bin width $\Delta\ell = 10$. The MCMC fitting of these spectra yields posterior distributions shown in the corner plots of Figure 2 (EE mode) and Figure 3 (BB mode). These plots illustrate the parameter constraints and correlations for the synchrotron and dust components.

5.2 Observational data

We perform three complementary analyses of the observational data to assess the robustness of our results and explore the scale dependence of the foreground components.

Power-law fitting with $\Delta\ell = 10$ We compute the angular power spectra from the real sky maps with bin width $\Delta\ell = 10$. The MCMC fitting of these spectra yields posterior distributions shown in the corner plots of Figure 4 (EE mode) and Figure 5 (BB mode). These plots illustrate the parameter constraints and correlations for the synchrotron and dust components derived from the observational data using the power-law model described in Section 4.

Power-law fitting with $\Delta\ell = 20$ To test the stability of our results with respect to binning choices, we repeat the analysis using a larger bin width of $\Delta\ell = 20$. The resulting posterior distributions are shown in Figure 6 (EE mode) and Figure 7 (BB mode). This wider binning reduces noise at the expense of spectral resolution and provides an independent check on the power-law model parameters.

Bin-to-bin analysis Finally, we perform a bin-to-bin analysis using bin width $\Delta\ell = 20$, where each ℓ bin is fitted independently, without assuming a power-law form for the multipole dependence. In this approach, only the synchrotron and dust amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), and the correlation coefficient (ρ) are fitted in each bin, while the multipole spectral index parameters α_s and α_d are not constrained. The results of this analysis are presented in Table 1 and visualized in Figure 8, which shows the fitted EE and BB parameter values with their uncertainties for each multipole bin, along with the reduced χ^2 values obtained for each bin. This approach allows us to identify potential deviations from the power-law assumption and assess the scale-dependent behavior of the foreground components.

References

Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641:A6.

Table 1: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	ℓ_{eff}	$A^{\mathrm{EE}}_{\mathrm{sync}}$ [μK^2]	$\beta^{\mathrm{EE}}_{\mathrm{sync}}$	$A^{\mathrm{EE}}_{\mathrm{dust}}$ [10^{-3} μK^2]	$\beta^{\mathrm{EE}}_{\mathrm{dust}}$	ρ^{EE}
30–49	39.0	9.00 ± 0.37	-3.15 ± 0.03	14.20 ± 0.12	1.47 ± 0.01	0.070 ± 0.010
50–69	59.0	2.46 ± 0.18	-3.18 ± 0.06	6.70 ± 0.06	1.52 ± 0.01	0.080 ± 0.014
70–89	79.0	0.99 ± 0.11	-3.21 ± 0.09	3.37 ± 0.03	1.52 ± 0.01	0.026 ± 0.018
90–109	99.0	0.50 ± 0.10	-3.27 ± 0.15	2.11 ± 0.02	1.51 ± 0.01	0.064 ± 0.091
110–129	119.0	0.19 ± 0.06	-3.18 ± 0.22	1.18 ± 0.02	1.47 ± 0.02	0.105 ± 0.045
130–149	139.0	0.12 ± 0.05	-2.82 ± 0.27	0.87 ± 0.01	1.43 ± 0.02	0.132 ± 0.048
150–169	159.0	0.16 ± 0.07	-3.48 ± 0.23	0.61 ± 0.01	1.51 ± 0.02	0.206 ± 0.095
170–189	179.0	0.07 ± 0.05	-3.14 ± 0.33	0.44 ± 0.01	1.54 ± 0.03	0.112 ± 0.165
ℓ range	ℓ_{eff}	$A^{\mathrm{BB}}_{\mathrm{sync}}$ [μK^2]	$\beta^{\mathrm{BB}}_{\mathrm{sync}}$	$A^{\mathrm{BB}}_{\mathrm{dust}}$ [10^{-3} μK^2]	$\beta^{\mathrm{BB}}_{\mathrm{dust}}$	ρ^{BB}
30–49	39.0	1.81 ± 0.22	-3.18 ± 0.09	13.05 ± 0.10	1.48 ± 0.01	0.114 ± 0.018
50–69	59.0	0.27 ± 0.08	-2.87 ± 0.18	5.02 ± 0.04	1.52 ± 0.01	0.141 ± 0.074
70–89	79.0	0.14 ± 0.08	-3.42 ± 0.25	2.76 ± 0.02	1.51 ± 0.01	0.073 ± 0.080
90–109	99.0	0.02 ± 0.02	-3.09 ± 0.27	1.44 ± 0.01	1.54 ± 0.01	0.256 ± 0.218
110–129	119.0	0.03 ± 0.03	-3.00 ± 0.27	1.05 ± 0.01	1.50 ± 0.02	0.164 ± 0.187
130–149	139.0	0.01 ± 0.03	-3.20 ± 0.29	0.71 ± 0.01	1.49 ± 0.02	0.004 ± 0.222
150–169	159.0	0.02 ± 0.03	-3.00 ± 0.32	0.47 ± 0.01	1.52 ± 0.02	0.103 ± 0.209
170–189	179.0	0.11 ± 0.06	-3.09 ± 0.51	0.41 ± 0.02	1.44 ± 0.09	0.027 ± 0.077

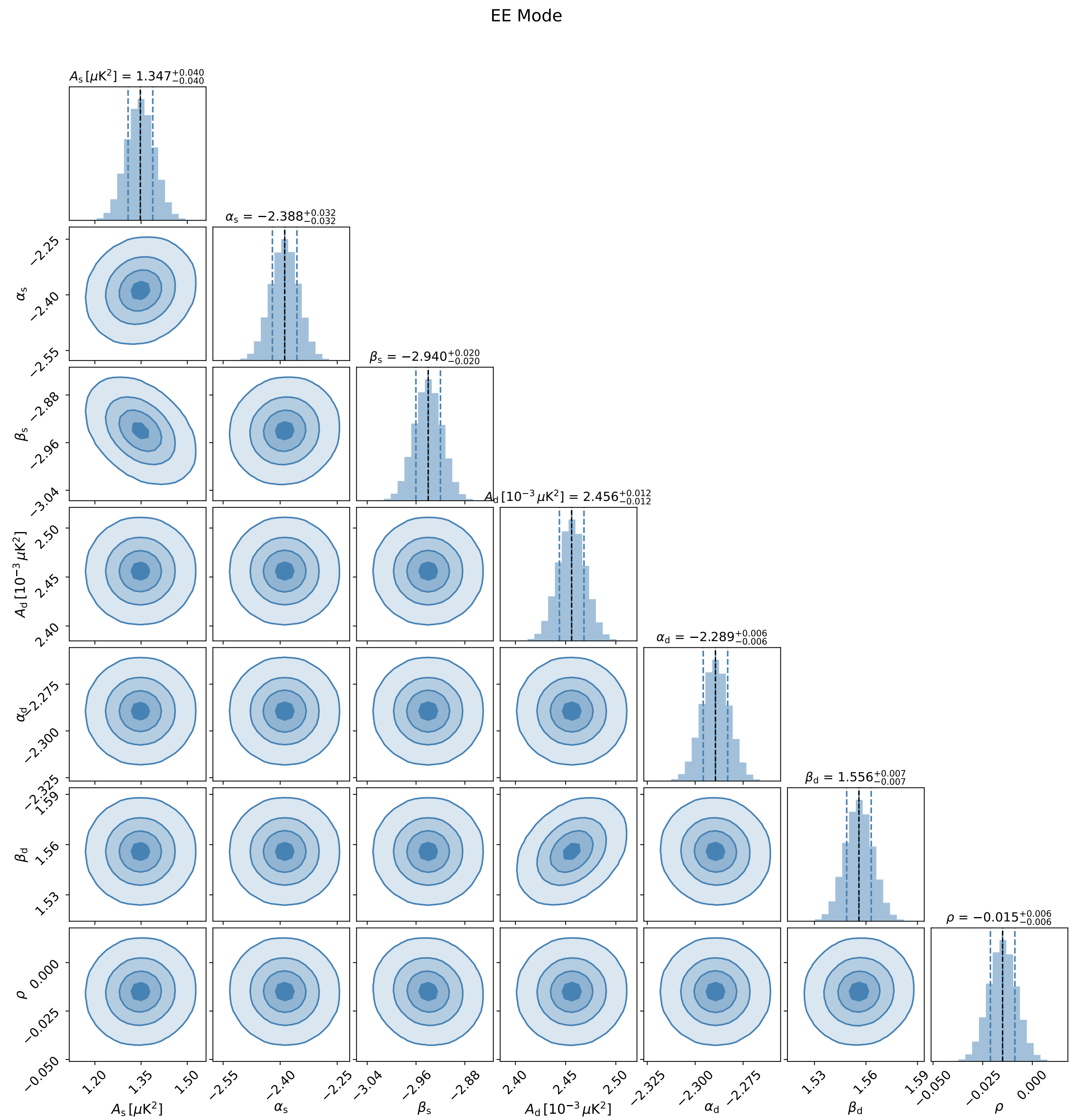

Figure 2: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of simulated EE mode data. The plot displays parameter constraints and correlations obtained from PySM-generated sky maps.

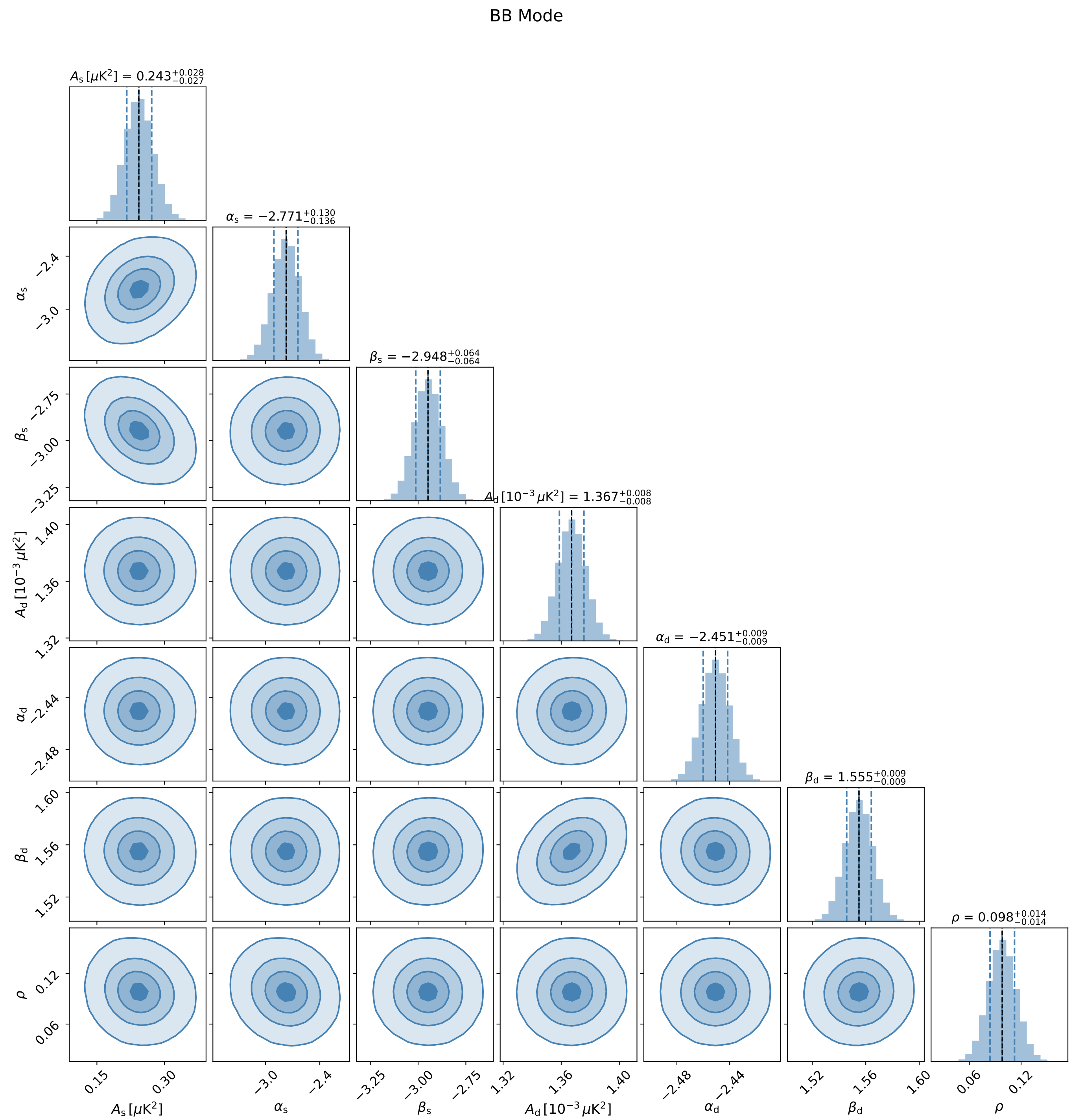

Figure 3: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of simulated BB mode data. The plot displays parameter constraints and correlations obtained from PySM-generated sky maps.

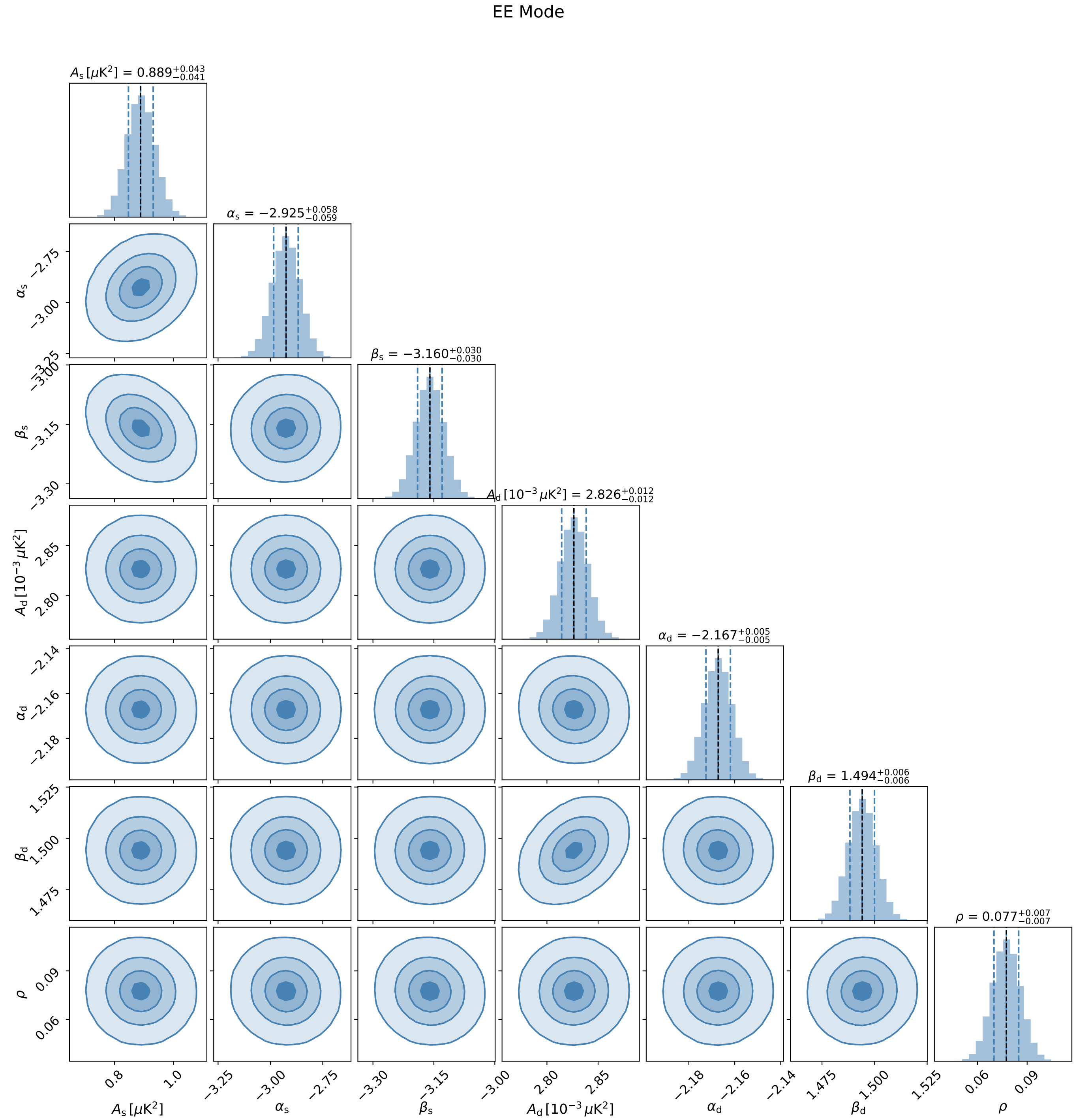

Figure 4: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational EE mode data with bin width $\Delta\ell = 10$. The plot displays parameter constraints and correlations obtained from power-law model fitting to the real sky maps.

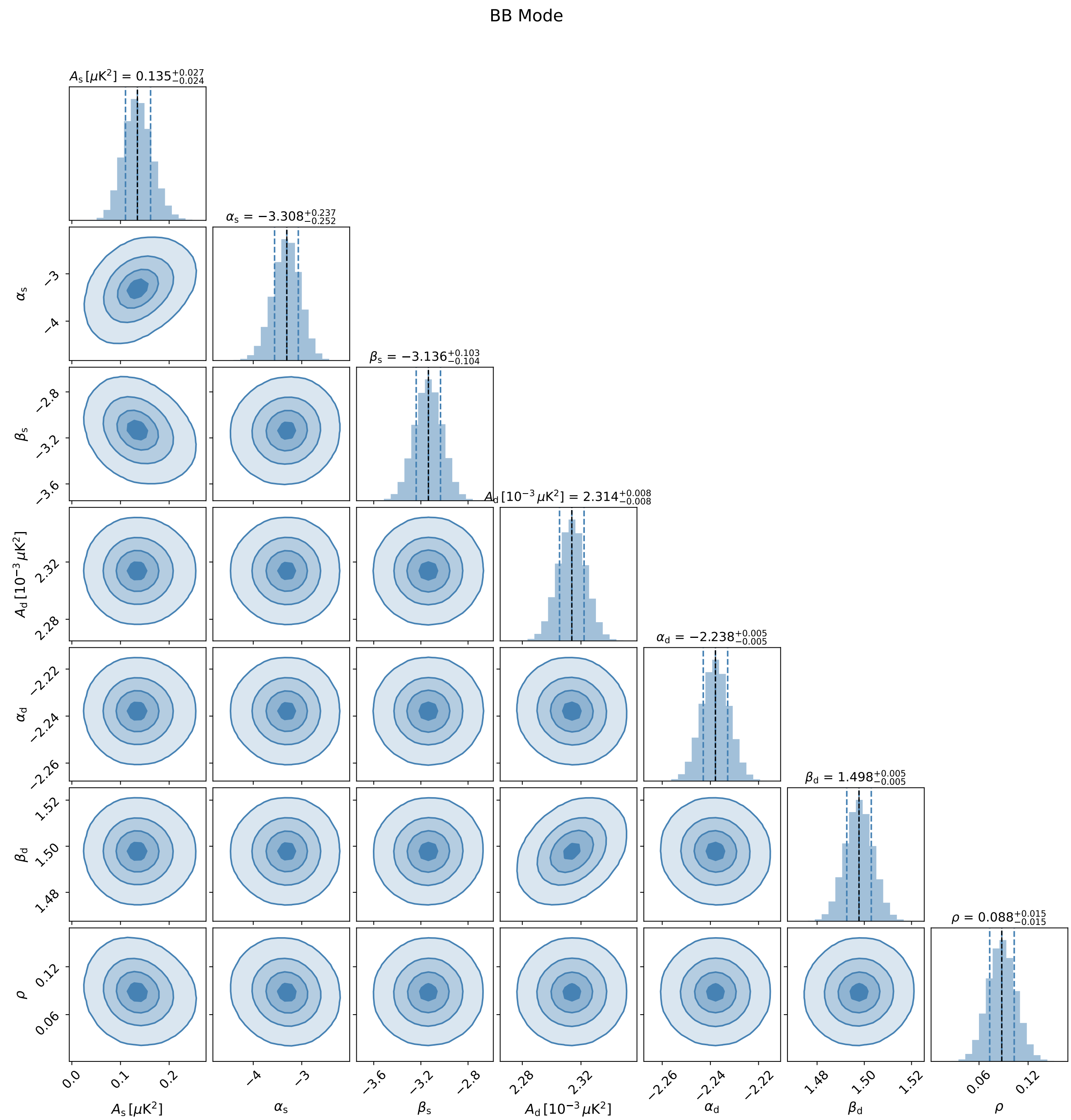

Figure 5: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational BB mode data with bin width $\Delta\ell = 10$. The plot displays parameter constraints and correlations obtained from power-law model fitting to the real sky maps.

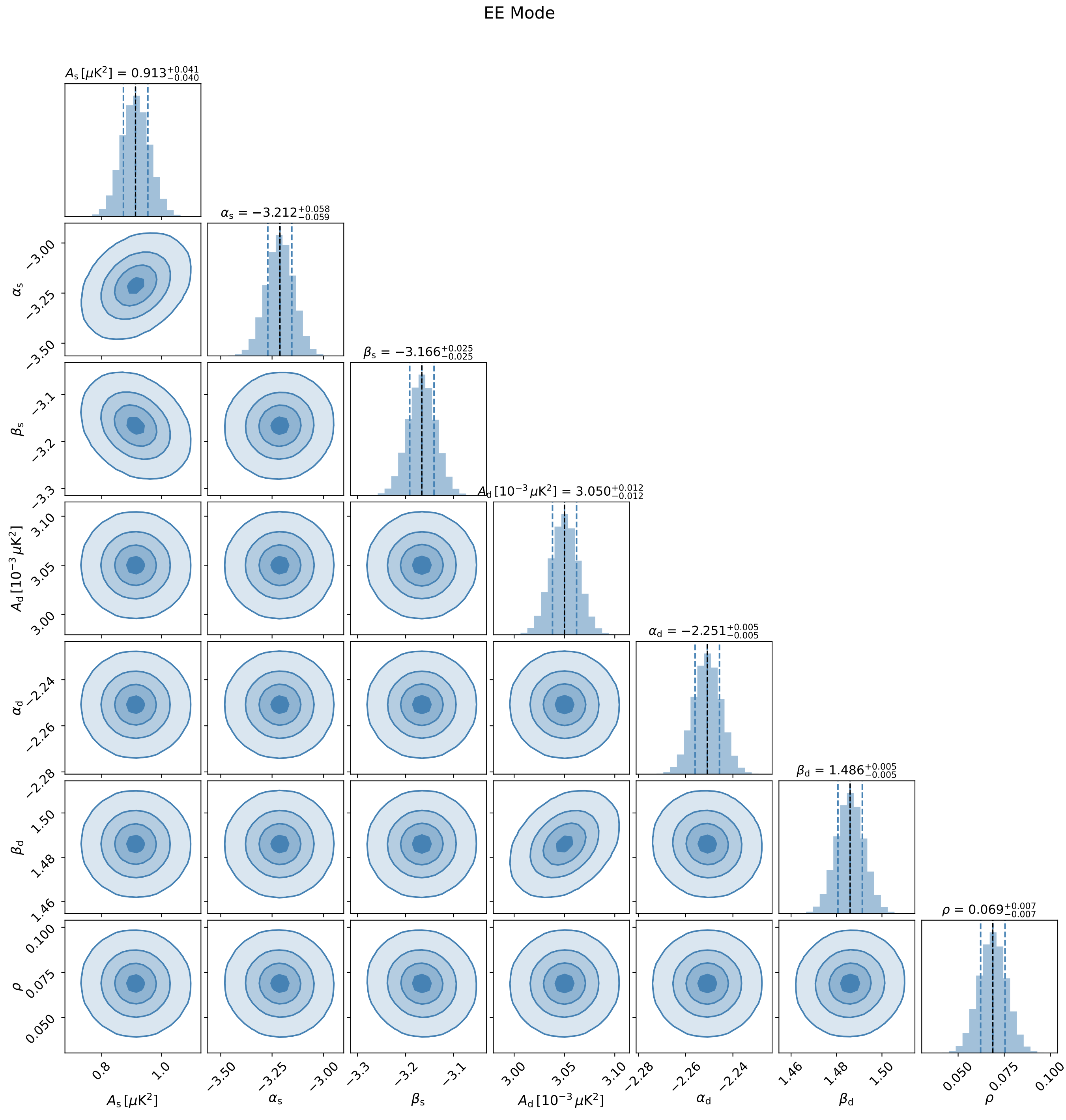

Figure 6: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational EE mode data with bin width $\Delta\ell = 20$. The plot displays parameter constraints and correlations obtained from power-law model fitting to the real sky maps with wider binning.

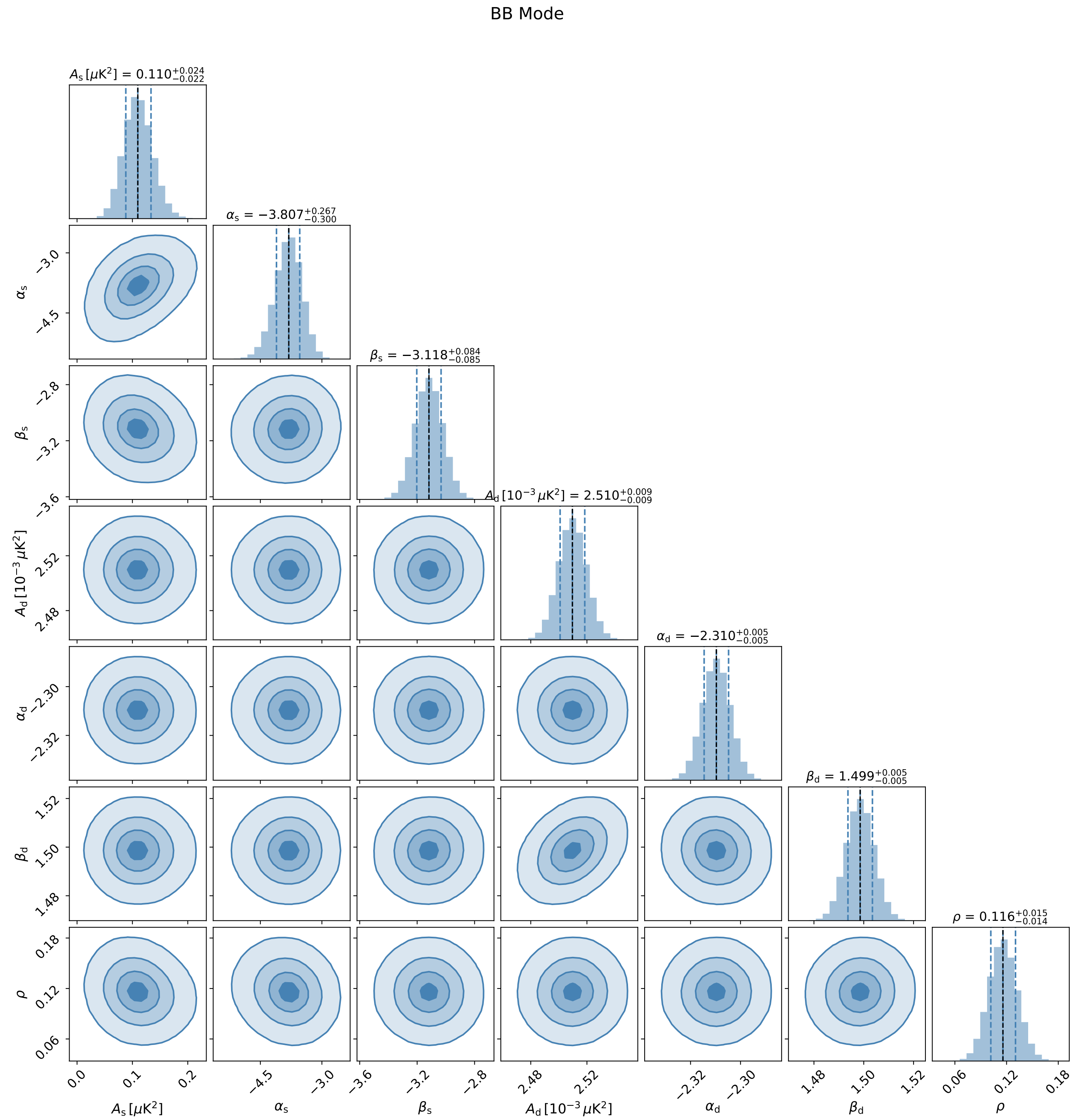

Figure 7: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational BB mode data with bin width $\Delta\ell = 20$. The plot displays parameter constraints and correlations obtained from power-law model fitting to the real sky maps with wider binning.

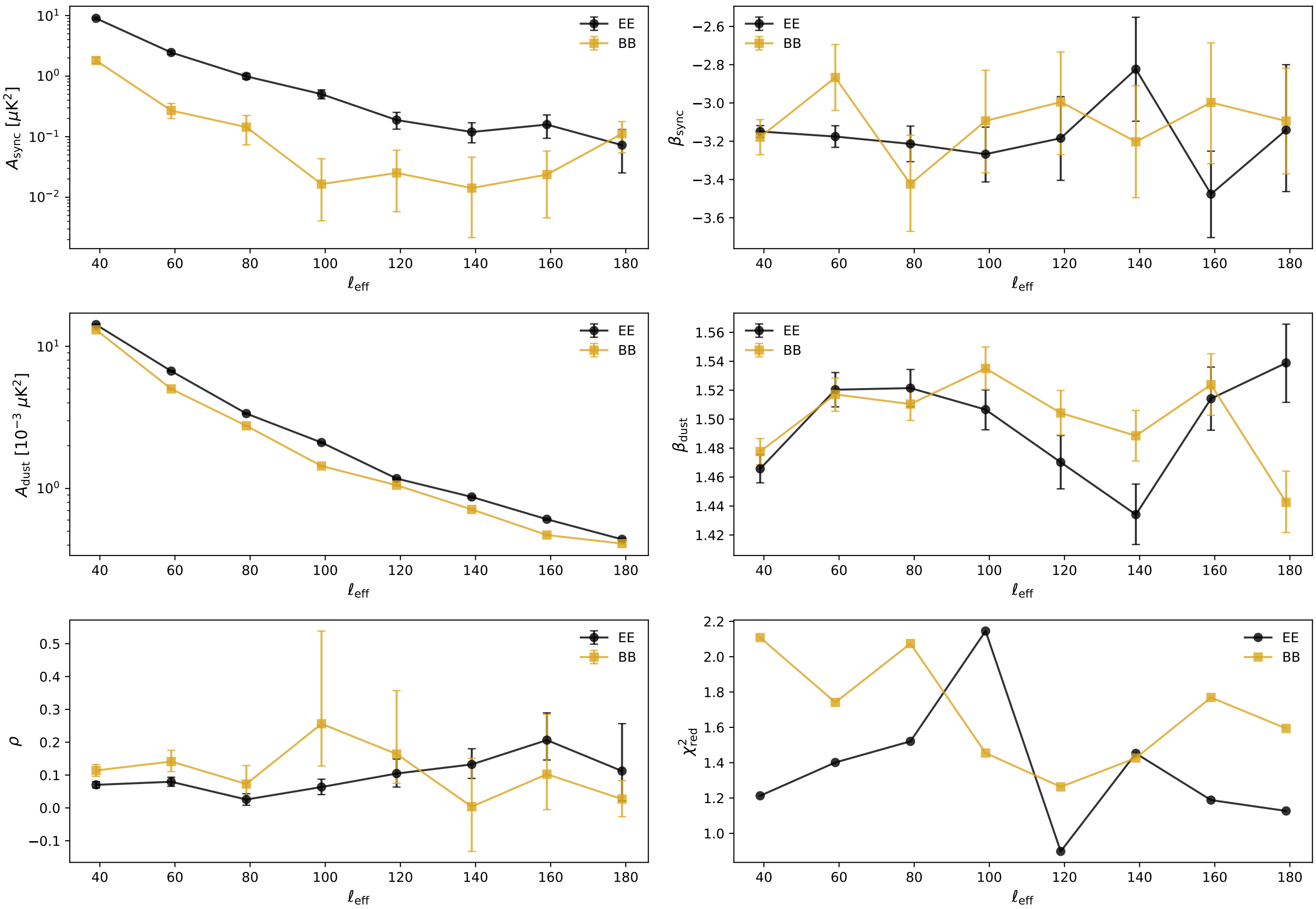

Figure 8: Bin-to-bin analysis results showing the evolution of fitted synchrotron and dust parameters as a function of multipole ℓ for both EE (top) and BB (bottom) modes. Each point represents an independent fit to a single multipole bin, with error bars indicating the 1σ uncertainties. The reduced χ^2 values for each bin are shown in the bottom panels, indicating the goodness of fit for the bin-to-bin analysis.